

Competitive Programming

Section 63: AtCoder Difficulty Ladder – Offline Problem Sheet

Competitive Programming Course

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1 Offline use

This sheet contains original concise briefs, not copied AtCoder statements. Each visible URL is the authoritative statement and submission page. Read it before submitting whenever you are online. Approximate ratings come from AtCoder Problems and are not official AtCoder labels.

2 Gate 1: short entry ramp

2.1 1. ABC083_B – Some Sums (approximately 100)

Official: https://atcoder.jp/contests/abc083/tasks/abc083_b?lang=en

Given N , A , and B , sum every integer x in $1..N$ whose decimal digit sum lies in the inclusive interval $[A, B]$. The bounds are small enough to inspect every candidate.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 1-2.

2.2 2. ABC210_C – Colorful Candies (approximately 400)

Official: https://atcoder.jp/contests/abc210/tasks/abc210_c?lang=en

Given N candy colors and a window length K , find the maximum number of distinct colors in any K consecutive positions.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 3, 6, and 12.

2.3 3. ABC167_D – Teleporter (approximately 800)

Official: https://atcoder.jp/contests/abc167/tasks/abc167_d?lang=en

A functional graph gives one next town for each town. Starting at town 1, report the town reached after exactly K moves, where K may be as large as 10^{18} .

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 16 and 27.

2.4 4. ABC130_D – Enough Array (approximately 900)

Official: https://atcoder.jp/contests/abc130/tasks/abc130_d?lang=en

For a positive array, count subarrays whose sum is at least K . N is large enough that enumerating both endpoints is too slow.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 4 and 6.

2.5 5. ABC138_D – Ki (approximately 900)

Official: https://atcoder.jp/contests/abc138/tasks/abc138_d?lang=en

Root a tree at vertex 1. Each query adds a value to one vertex and every descendant of that vertex. Print the final value at every vertex.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 16 and 26.

3 Gate 2: core intermediate

3.1 6. ABC218_E – Destruction (approximately 1000)

Official: https://atcoder.jp/contests/abc218/tasks/abc218_e?lang=en

Delete edges from a connected weighted undirected graph while keeping it connected, maximizing the sum of deleted edge weights. Weights may be negative.

Constraints and exact input/output syntax: see the official link. Course bridge: Section 18.

3.2 7. ABC075_C – Bridge (approximately 1100)

Official: https://atcoder.jp/contests/abc075/tasks/abc075_c?lang=en

In a small connected undirected graph, count edges whose individual removal disconnects the graph.

Constraints and exact input/output syntax: see the official link. Course bridge: Section 16.

3.3 8. ABC180_E – Traveling Salesman among Aerial Cities (approximately 1300)

Official: https://atcoder.jp/contests/abc180/tasks/abc180_e?lang=en

Visit every 3D city, starting and ending at city 1. Travel cost is $|dx|+|dy|+\max(0, z_{\text{destination}}-z_{\text{source}})$, so direction matters.

Constraints and exact input/output syntax: see the official link. Course bridge: Section 27.

3.4 9. ABC137_D – Summer Vacation (approximately 1300)

Official: https://atcoder.jp/contests/abc137/tasks/abc137_d?lang=en

There are M working days. Job i becomes available after A_i days and pays B_i ; at most one job can be completed per day. Maximize reward.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 9 and 39.

3.5 10. ABC135_D – Digits Parade (approximately 1300)

Official: https://atcoder.jp/contests/abc135/tasks/abc135_d?lang=en

Replace every ‘?’ in a decimal string by a digit. Count replacements whose resulting integer is congruent to 5 modulo 13, modulo $1e9+7$.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 22 and 31.

3.6 11. ABC187_E – Through Path (approximately 1400)

Official: https://atcoder.jp/contests/abc187/tasks/abc187_e?lang=en

Each query names a tree edge, chooses one of its two sides, and adds x to every vertex on that side. Print all final vertex values.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 4 and 26.

3.7 12. ABC184_E – Third Avenue (approximately 1400)

Official: https://atcoder.jp/contests/abc184/tasks/abc184_e?lang=en

Find a shortest path through a grid with walls. Equal lowercase letters are teleport stations: from a letter cell one move may reach any other cell with that letter.

Constraints and exact input/output syntax: see the official link. Course bridge: Section 15.

3.8 13. ABC245_F – Endless Walk (approximately 1500)

Official: https://atcoder.jp/contests/abc245/tasks/abc245_f?lang=en

Count vertices of a directed graph from which one can follow directed edges forever.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 17 and 38.

3.9 14. ABC174_F – Range Set Query (approximately 1500)

Official: https://atcoder.jp/contests/abc174/tasks/abc174_f?lang=en

Answer many inclusive array-range queries asking for the number of distinct colors.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 13 and 36.

3.10 15. ABC210_D – National Railway (approximately 1500)

Official: https://atcoder.jp/contests/abc210/tasks/abc210_d?lang=en

Choose two distinct grid cells for stations. Their cost is both cell prices plus C times Manhattan distance; minimize it.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 22 and 28.

4 Gate 3: strong intermediate

4.1 16. ABC190_E – Magical Ornament (approximately 1600)

Official: https://atcoder.jp/contests/abc190/tasks/abc190_e?lang=en

In an unweighted graph, find the minimum number of visited vertices in a walk that visits every one of K special vertices, or report impossibility.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 15 and 27.

4.2 17. ABC196_E – Filters (approximately 1700)

Official: https://atcoder.jp/contests/abc196/tasks/abc196_e?lang=en

Compose many operations of three kinds on a number: add a , replace by $\max(x, a)$, or replace by $\min(x, a)$. Answer the final value for many inputs.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 10 and 49.

4.3 18. ABC204_E – Rush Hour 2 (approximately 1700)

Official: https://atcoder.jp/contests/abc204/tasks/abc204_e?lang=en

An undirected edge entered at integer time t arrives at $t + C + \text{floor}(D/(t+1))$; waiting before entering is allowed. Find the earliest arrival from 1 to N .

Constraints and exact input/output syntax: see the official link. Course bridge: Section 19.

4.4 19. ABC192_F – Potion (approximately 1800)

Official: https://atcoder.jp/contests/abc192/tasks/abc192_f?lang=en

Choose a nonempty subset of the given positive values so that adding the same integer amount to every chosen value makes their sum equal X . Minimize that common added amount.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 23 and 27.

4.5 20. ABC061_D – Score Attack (approximately 1800)

Official: https://atcoder.jp/contests/abc061/tasks/abc061_d?lang=en

Find the maximum score of a directed path from vertex 1 to vertex N . If a positive cycle usable on such a path makes the score unbounded, print `inf`.

Constraints and exact input/output syntax: see the official link. Course bridge: Section 19.

4.6 21. ABC339_G – Smaller Sum (approximately 1800)

Official: https://atcoder.jp/contests/abc339/tasks/abc339_g?lang=en

For online-decoded queries, return the sum of array values no greater than X inside $[L,R]$. Each query's parameters are XORed with the previous answer.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 13 and 42.

4.7 22. ABC172_E – NEQ (approximately 1900)

Official: https://atcoder.jp/contests/abc172/tasks/abc172_e?lang=en

Count ordered pairs of length- N injective sequences A and B over $1..M$ such that A_i differs from B_i at every position, modulo $1e9+7$.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 32 and 33.

4.8 23. ABC285_F – Substring of Sorted String (approximately 1900)

Official: https://atcoder.jp/contests/abc285/tasks/abc285_f?lang=en

Support point character updates and queries asking whether a substring could appear as one contiguous block in the globally sorted string.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 36, 37, and 44.

4.9 24. ABC223_F – Parenthesis Checking (approximately 1900)

Official: https://atcoder.jp/contests/abc223/tasks/abc223_f?lang=en

Support swaps of two parentheses and queries asking whether an inclusive substring is a correct bracket sequence.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 4 and 37.

4.10 25. ABC254_F – Rectangle GCD (approximately 1900)

Official: https://atcoder.jp/contests/abc254/tasks/abc254_f?lang=en

For arrays A and B , the conceptual matrix has value A_i+B_j . Answer the gcd of every value in many queried row-column rectangles.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 13 and 35.

4.11 26. ABC253_F – Operations on a Matrix (approximately 1900)

Official: https://atcoder.jp/contests/abc253/tasks/abc253_f?lang=en

An initially zero matrix supports adding x to every cell in a column interval, assigning every cell of one row to x , and point-value queries.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 36 and 42.

4.12 27. ABC199_F – Graph Smoothing (approximately 2100)

Official: https://atcoder.jp/contests/abc199/tasks/abc199_f?lang=en

Repeat a random edge-based averaging operation K times on graph vertex values and print the expected final value at every vertex modulo a prime.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 32 and 50.

4.13 28. ABC256_F – Cumulative Cumulative Cumulative Sum (approximately 2100)

Official: https://atcoder.jp/contests/abc256/tasks/abc256_f?lang=en

Support point assignments in an array and queries for a third-order prefix sum, all modulo 998244353.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 4 and 36.

4.14 29. ABC205_F – Grid and Tokens (approximately 2100)

Official: https://atcoder.jp/contests/abc205/tasks/abc205_f?lang=en

Each token may be placed in one cell of its allowed rectangle or not placed. No two placed tokens may share a row or a column; maximize the number placed.

Constraints and exact input/output syntax: see the official link. Course bridge: Section 45.

4.15 30. ABC237_G – Range Sort Query (approximately 2100)

Official: https://atcoder.jp/contests/abc237/tasks/abc237_g?lang=en

A permutation undergoes many operations sorting a subarray ascending or descending. Report the final position of a distinguished value X .

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 8 and 37.

5 Gate 4: advanced

5.1 31. ABC132_F – Small Products (approximately 2100)

Official: https://atcoder.jp/contests/abc132/tasks/abc132_f?lang=en

Count length-K positive integer sequences satisfying the problem's adjacent product bound, modulo $1e9+7$, with N too large for one state per value.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 22 and 49.

5.2 32. ABC296_G – Polygon and Points (approximately 2200)

Official: https://atcoder.jp/contests/abc296/tasks/abc296_g?lang=en

For many query points, classify each as strictly inside, on the boundary of, or outside a convex polygon listed in cyclic order.

Constraints and exact input/output syntax: see the official link. Course bridge: Section 54.

5.3 33. ABC239_G – Builder Takahashi (approximately 2200)

Official: https://atcoder.jp/contests/abc239/tasks/abc239_g?lang=en

Vertices other than 1 and N have destruction costs. Find the minimum cost of vertices whose removal disconnects 1 from N, and output one optimal set.

Constraints and exact input/output syntax: see the official link. Course bridge: Section 45.

5.4 34. ABC238_G – Cubic? (approximately 2300)

Official: https://atcoder.jp/contests/abc238/tasks/abc238_g?lang=en

For each array range, decide whether the product of its elements is a perfect cube.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 33 and 42.

5.5 35. ABC274_G – Security Camera 3 (approximately 2300)

Official: https://atcoder.jp/contests/abc274/tasks/abc274_g?lang=en

Cover all open grid cells using the minimum number of axis-aligned camera segments, with walls breaking visibility.

Constraints and exact input/output syntax: see the official link. Course bridge: Section 45.

5.6 36. ABC265_G – 012 Inversion (approximately 2400)

Official: https://atcoder.jp/contests/abc265/tasks/abc265_g?lang=en

An array over $\{0,1,2\}$ supports range symbol permutations and queries for the inversion count of a range.

Constraints and exact input/output syntax: see the official link. Course bridge: Section 37.

5.7 37. ABC338_G – eval (approximately 2600)

Official: https://atcoder.jp/contests/abc338/tasks/abc338_g?lang=en

Sum the usual arithmetic value of every digit-bounded substring of a long expression containing digits, '+', and '*', modulo 998244353.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 31 and 49.

5.8 38. AGC003_D – Anticube (approximately 2600)

Official: https://atcoder.jp/contests/abc003/tasks/abc003_d?lang=en

Choose the largest subset of integers such that the product of no two chosen elements is a perfect cube.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 33 and 35.

5.9 39. ABC354_G – Select Strings (approximately 2700)

Official: https://atcoder.jp/contests/abc354/tasks/abc354_g?lang=en

Select a maximum-value compatible subset of weighted strings under the problem's substring-conflict rule.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 44 and 45.

6 Gate 5: orange

6.1 40. AGC026_E – Synchronized Subsequence (approximately 3000)

Official: https://atcoder.jp/contests/agc026/tasks/agc026_e?lang=en

A length- $2N$ string contains N ‘a’ and N ‘b’. For every rank i , either select both the i -th ‘a’ and i -th ‘b’, or neither; maximize the selected subsequence lexicographically.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 22 and 44.

6.2 41. AGC006_D – Median Pyramid Hard (approximately 3100)

Official: https://atcoder.jp/contests/agc006/tasks/agc006_d?lang=en

The bottom row is a permutation of $1..2N-1$. Each upper block is the median of its three children; find the value at the pyramid’s top.

Constraints and exact input/output syntax: see the official link. Course bridge: Section 8.

6.3 42. ARC108_E – Random IS (approximately 3100)

Official: https://atcoder.jp/contests/arc108/tasks/arc108_e?lang=en

Repeatedly choose uniformly among currently addable chairs while keeping marked chair IDs increasing left-to-right. Find the expected final number marked.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 25 and 50.

6.4 43. AGC023_F – 01 on Tree (approximately 3100)

Official: https://atcoder.jp/contests/agc023/tasks/agc023_f?lang=en

Linearize a rooted tree so every ancestor precedes its descendants. Vertex labels are 0 or 1; minimize inversions in the resulting binary sequence.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 9, 26, and 39.

6.5 44. AGC023_D – Go Home (approximately 3200)

Official: https://atcoder.jp/contests/agc023/tasks/agc023_d?lang=en

A bus starts at S carrying populations bound for apartments on a line. At every unit step, rational passengers vote for the direction that gets them home sooner; find when the final passenger exits.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 9 and 11.

6.6 45. ARC069_D – Flags (approximately 3200)

Official: https://atcoder.jp/contests/arc069/tasks/arc069_d?lang=en

Flag i may be placed at either x_i or y_i . Maximize the minimum distance between every pair of placed flags.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 8, 17, and 37.

7 Gate 6: red

7.1 46. AGC017_F – Zigzag (approximately 3400)

Official: https://atcoder.jp/contests/agc017/tasks/agc017_f?lang=en

Count ordered noncrossing down-left/down-right paths through a triangular grid, subject to fixed direction choices on selected path-step pairs.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 27 and 37.

7.2 47. AGC032_E – Modulo Pairing (approximately 3400)

Official: https://atcoder.jp/contests/agc032/tasks/agc032_e?lang=en

Pair $2N$ residues modulo M to minimize the maximum value of $(x+y) \bmod M$ among the N pairs.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 8 and 9.

7.3 48. AGC005_F – Many Easy Problems (approximately 3400)

Official: https://atcoder.jp/contests/agc005/tasks/agc005_f?lang=en

For every $K=1..N$, sum over all K -vertex sets the number of vertices in their minimum containing subtree, modulo 924844033.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 26, 32, and 51.

7.4 49. AGC024_F – Simple Subsequence Problem (approximately 3500)

Official: https://atcoder.jp/contests/agc024/tasks/agc024_f?lang=en

A compact bit-table describes a set of binary strings of length at most N . Find the longest, then lexicographically smallest, string that is a subsequence of at least K members of the set.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 27 and 44.

7.5 50. AGC016_F – Games on DAG (approximately 3800)

Official: https://atcoder.jp/contests/agc016/tasks/agc016_f?lang=en

From a forward-edge DAG, choose any subset of edges. Two impartial-game tokens start at vertices 1 and 2; count edge subsets for which the first player wins.

Constraints and exact input/output syntax: see the official link. Course bridge: Sections 27, 29, and 33.